The minimum number of edges in a pancyclic graph and the number of cycles in a graph

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Abstract

Let $n + h(n)$ be the least number of edges of a pancyclic graph on n vertices, that is, of a graph that contains a cycle of every length from 3 to n. KNT recently proved that $h(n) = \log_2 n + \log^* n + O(1)$, as Erdős had asked. The proof rests on one estimate: if a small union of cycles W is removed from a connected graph of large cycle rank, then a uniformly random even subgraph, restricted to the remaining edges, is a forest with probability at most $\frac{1}{2} + o(1)$.

We give a compact account of the theorem, organised around this estimate and a single counting lemma, and record what else the two of them give. First, a graph of cycle rank r has at most $(\frac{1}{2} + o(1))2^r$ cycles. This answers, asymptotically, a question of Entringer and Slater from 1981; it was known for planar graphs and for graphs of average degree slightly above 4, while for cubic graphs the best bound was $\frac{15}{16}2^r$. Second, we describe the regime in which the counting lemma loses nothing: it cannot last for more than $|V(W)| + 3$ ranks beyond W, and a two-terminal graph that carries k binary digits of path lengths without loss has no path between its terminals with fewer than k edges. These elementary facts account for the shape of the extremal construction and show where the iterated logarithm comes from.

1 Introduction

A graph on n vertices is *pancyclic* if it contains a cycle of every length $3, 4, \dots, n$. Let $n + h(n)$ be the least number of edges of a pancyclic graph on n vertices. Throughout, log is the binary logarithm, $\log^{(j)}$ is its j-fold iterate, $\log^* x$ is the least $j \geq 0$ with $\log^{(j)} x \leq 1$, and ln is the natural logarithm. Graphs are finite and, unless we say otherwise, simple.

Theorem 1.1 (KNT [11])**.** *There is an absolute constant A such that, for every $n \geq 3$,*

$$\log n + \log^* n - A \;\leq\; h(n) \;\leq\; \lfloor \log n \rfloor + \log^* n + 1.$$

Bondy [5] stated without proof that $\log(n-1) - 1 \leq h(n) \leq \log n + \log^* n + O(1)$. The lower bound is a count: a connected graph with $n + h$ edges has at most $2^{h+1} - 1$ cycles, and a pancyclic graph needs $n - 2$ of them. A proof of it was written down by Griffin [9], and the first published proof of the upper bound appears to be the one in the book of George, Khodkar and Wallis [8, §4.5]; see [4]. Erdős believed that the upper bound is closer to the truth, but could not even show that $h(n) - \log n \to \infty$; this is Problem #1016 in Bloom's collection of Erdős problems. Theorem 1.1 confirms his belief.

Two ingredients. For a graph G let $\mathcal{Z}(G)$ be its cycle space over $\mathbb{F}_2$: the set of *even subgraphs*, that is, of edge sets that meet every vertex an even number of times, with symmetric difference as addition. Its dimension is the *cycle rank* $\beta(G) = |E(G)| - |V(G)| + \mathrm{comp}(G)$, where comp counts components. A *cycle* is a connected 2-regular subgraph, and we do not distinguish between a subgraph without isolated vertices and its edge set.

The count above uses only that the cycles of a graph are distinct elements of $\mathcal{Z}(G)$. To use that their lengths fill an interval, fix a subgraph $W \subseteq G$ and sort the even subgraphs ξ of G by their *outside part* $\xi \setminus E(W)$. If $\beta(G) = r$, $\beta(W) = t$ and W has m edges, there are 2^{r-t} possible outside parts, and cycles with the same outside part differ only inside W, so they have at most m different lengths. Hence G has cycles of at most $m\,2^{r-t}$ lengths. When W is a union of cycles of lengths $3, 4, 5, \ldots$ this can be nearly attained, as long as r is not much larger than t. The theorem that drives everything says that it is off by a factor of two once r is large: the outside part of a cycle that meets W is a forest, and only half of the outside parts are forests.

Theorem 1.2 (Forest estimate, KNT [11])**.** *There is r_0 such that the following holds for all $r \geq r_0$. Let G be a connected simple graph with $\beta(G) = r$, and let $W \subseteq G$ be a nonempty union of cycles with at most $\sqrt{r}$ edges and at most $(\log r)/100$ components. If ξ is a uniformly random element of $\mathcal{Z}(G)$, then*

$$\mathbb{P}(\xi \setminus E(W) \text{ is a forest}) \;\leq\; \frac{1}{2} + \frac{20}{\log^{(3)} r}. \tag{1}$$

The constant $\frac{1}{2}$ is best possible (Remark 2.2). Section 3 derives the lower bound of Theorem 1.1: a witness W of rank about $2R$ that contains the short cycles has few edges, so (1) applies to every graph of rank 2^{256R}, and the quantity $\max(\text{number of lengths})/2^{\text{rank}}$ halves each time the rank passes one exponential level. Section 4 gives the construction and Section 6 proves Theorem 1.2.

The number of cycles. Let $\psi(r)$ be the largest number of cycles in a graph of cycle rank r. Entringer and Slater [6] observed that $\psi(r) \leq 2^r - 1$, that the maximum is attained by cubic graphs and that $\psi(r) > 2^{r-1}$, and they asked whether 2^{r-1} is essentially the truth; this is usually quoted as the conjecture $\psi(r) \sim 2^{r-1}$. Aldred and Thomassen [2] proved $\psi(r) \leq \frac{15}{16}2^r$, and $2^{r-1} + o(2^{r-1})$ for planar graphs. Alahmadi, Aldred, dela Cruz, Solé and Thomassen [1] showed that a graph with p vertices and $q > 2p + 3\log(3p)$ edges has at most 2^{q-p} cycles. This left the sparse graphs, and in particular the cubic ones, for which $\frac{15}{16}2^r$ remained the best bound that we are aware of [10]. The forest estimate closes this gap.

Theorem 1.3. *For all sufficiently large r,*

$$\psi(r) \;\leq\; \Big(\frac{1}{2} + \frac{21}{\log^{(3)} r}\Big)2^r.$$

In particular $\psi(r) \sim 2^{r-1}$.

Exact counting. The factor $\frac{1}{2}$ is forced only when the rank is large. Our second complement describes the opposite regime, in which nothing is lost at all.

Theorem 1.4. (a) *Let N be a connected graph and let $s \neq t$ be two of its vertices. Then N has at most $2^{\beta(N)}$ paths from s to t. If it has exactly $2^{\beta(N)}$, then each of these paths has at least $\beta(N)$ edges, and deleting the edges of any one of them leaves a forest.*

(b) *Let G be a connected graph and let $W \subseteq G$ be a subgraph that contains a cycle. If every coset of $\mathcal{Z}(W)$ in $\mathcal{Z}(G)$ contains a cycle, then $G - E(W)$ has no two edge-disjoint cycles, and $\beta(G) \le \beta(W) + |V(W)| + 3$.*

Part (b) says that the count described above can be exact for at most $|V(W)| + 3$ ranks beyond W. Part (a) says that a gadget which turns k units of rank into 2^k path lengths, like the necklace in the construction of Section 4, has no path between its terminals with fewer than k edges. The cycles through it are therefore long, the short cycle lengths have to come from a second, exponentially smaller gadget, and so on: this is where the iterated logarithm comes from in the construction. The lower bound of Theorem 1.1 says that allowing losses does not help by more than a constant. Section 7 also contains a stability statement of the same elementary kind (Proposition 7.4).

What is new. Theorems 1.1 and 1.2 and the strategy of their proofs are due to KNT [11], whose argument has been formalised in Lean 4. Sections 3, 4 and 6 follow [11]. The changes there are expository: one counting lemma (Lemma 2.1) serves both applications, the link formula is proved without contractions, and the two facts needed about the non-backtracking matrix are derived directly. Theorems 1.3 and 1.4 and Proposition 7.4 are new as far as we know. They have not been formalised; as a check, the elementary statements of Sections 2, 4 and 7 and the lemmas of Section 6 on links and on the non-backtracking matrix were tested by computer on small graphs.

2 Counting by cosets

Let W be a subgraph of a graph G, and put $r = \beta(G)$, $t = \beta(W)$ and $m = |E(W)|$. The map $\xi \mapsto \xi \setminus E(W)$ is linear on $\mathcal{Z}(G)$ with kernel $\mathcal{Z}(W)$. We call the elements of its image $\mathcal{X} = \mathcal{X}(G, W)$ *words*. There are 2^{r-t} words, each word is the outside part of exactly 2^t even subgraphs, and if ξ is uniform on $\mathcal{Z}(G)$ then $\xi \setminus E(W)$ is uniform on $\mathcal{X}$. Let

$$p(G, W) = \mathbb{P}\big(\xi \setminus E(W) \text{ is a forest}\big)$$

be the proportion of words that are forests.

Lemma 2.1 (Counting lemma)**.** *With this notation and $p = p(G, W)$,*

(a) *G has at most $2^r p + 2^{r-t}(1-p) \le 2^r\big(p + 2^{-t}\big)$ cycles;*

(b) *the cycles of G have at most $2^{r-t}(pm + 1 - p)$ different lengths.*

Proof. Let C be a cycle of G with word $x = C \setminus E(W)$. If x is not a forest, it contains a cycle, and since $x \subseteq C$ this forces $C = x$; so a word that is not a forest is the outside part of at most one cycle. If x is a forest, it is the outside part of at most 2^t cycles C, and $C \cap E(W) \neq \emptyset$ for each of them, because $C \neq x$ when $x \neq \emptyset$ and $C \subseteq W$ when $x = \emptyset$. So their lengths $|x| + |C \cap E(W)|$ take at most m values. Now sum over the $p\,2^{r-t}$ words that are forests and the $(1-p)2^{r-t}$ words that are not. □

Remark 2.2 (Two points of attachment). Suppose that $G = W \cup N$, where W and N are connected, have no common edge, and have exactly two common vertices s and t. An $\{s, t\}$-*join* of N is an edge set in which s and t have odd degree and all other vertices have even degree.

The words are the $2^{\beta(N)}$ even subgraphs of N and the $2^{\beta(N)}$ joins of N, and a nonempty even subgraph is not a forest. Hence

$$p(G,W) \;\le\; \tfrac{1}{2} + 2^{-(r-t)}.$$

So (1) is trivial, and holds without the error term, for a subgraph attached at two vertices. Equality holds when every join of N is a path, for instance when N is a chain of k four-cycles joined at opposite vertices and s, t are its two ends. With W a triangle this example also shows that the constant $\frac{1}{2}$ in (1) cannot be lowered.

3 The lower bound

For an integer $R \ge 1$ let $\Phi(R)$ be the supremum of $(L-2)\,2^{-\beta(G)}$ over all simple graphs G with $\beta(G) \ge R$ and all integers $L \ge 2$ such that G contains a cycle of each length $3, \dots, L$. Since a graph of rank β has at most $2^\beta - 1$ cycles, $\Phi(R) \le 1$, and Φ is non-increasing.

Lemma 3.1 (Witness)**.** *Let $R \ge 1$ and $\varphi = \Phi(R)$, and let G contain a cycle of each length $3, \dots, L$, where $L \ge 2^{2R}$. Then G has a subgraph W that is a union of at most $2R$ cycles, with $\beta(W) \ge 2R$ and $|E(W)| \le \varphi\, 2^{2R} + 2^R + 5R$.*

Proof. Put $W_0 = \emptyset$ and $W_{i+1} = W_i \cup C_i$, where C_i is a cycle of G whose length ℓ_i is the least integer ≥ 3 that is not the length of a cycle of W_i; stop as soon as the rank is at least $2R$. Let $s = \beta(W_i) < 2R$. The graph W_i contains cycles of the $\ell_i - 3$ lengths $3, \dots, \ell_i - 1$. Therefore $\ell_i - 3 \le 2^s - 1$, which gives $\ell_i \le 2^{2R-1} + 2 \le L$, so that C_i exists; and $\ell_i - 3 \le \varphi\, 2^s$ if $s \ge R$, by the definition of Φ. As C_i is not a subgraph of W_i, the rank increases at every step. So there are at most $2R$ steps, their ranks s are distinct, and

$$|E(W)| \;\le\; \sum_i \ell_i \;\le\; \sum_{s=0}^{R-1} (2^s + 2) + \sum_{s=R}^{2R-1} (\varphi\, 2^s + 3) \;\le\; \varphi\, 2^{2R} + 2^R + 5R. \qquad \square$$

Proposition 3.2 (Near-halving)**.** *For every sufficiently large integer R,*

$$\Phi(2^{256R}) \;\le\; \Big(\frac{1}{2} + \frac{20}{\log^{(2)} R}\Big)\Phi(R) + 2^{2-R}.$$

Proof. Let G be a simple graph of rank $r \ge 2^{256R}$ with a cycle of each length $3, \dots, L$. Identifying a vertex of one component with a vertex of another changes neither the rank nor the set of cycles, so we may assume that G is connected. If $L < 2^{2R}$ then $(L-2)2^{-r} < 2^{2-R}$. Otherwise let W be as in Lemma 3.1, with m edges and rank $t \ge 2R$. Then $m \le 2^{2R+1} \le \sqrt{r}$ and $\mathrm{comp}(W) \le 2R \le (\log r)/100$, so Theorem 1.2 gives $p(G,W) \le \frac{1}{2} + 20/\log^{(3)} r \le \frac{1}{2} + \delta$ with $\delta = 20/\log^{(2)} R$; here we used $\log^{(3)} r \ge \log(8 + \log R) \ge \log^{(2)} R$. By Lemma 2.1(b),

$$\frac{L-2}{2^r} \;\le\; \frac{pm+1}{2^t} \;\le\; \Big(\frac{1}{2} + \delta\Big)\big(\varphi + 2^{-R} + 5R\,2^{-2R}\big) + 2^{-2R} \;\le\; \Big(\frac{1}{2} + \delta\Big)\varphi + 2^{2-R}. \qquad \square$$

Corollary 3.3. *There is a constant c such that $\Phi(r) \le c\, 2^{-\log^* r}$ for all $r \ge 1$.*

Proof. Fix $R_0 \ge 2^{18}$ so large that Proposition 3.2 holds for all $R \ge R_0$, and let $R_{j+1} = 2^{256R_j}$ and $\delta_j = 20/\log^{(2)} R_j$. For $j \ge 2$ we have $\log^{(2)} R_j = 8 + 256R_{j-2}$, so $\sum_j \delta_j < \infty$; also $\sum_j 2^{j-R_j} < \infty$.

The numbers $a_j = 2^j \Phi(R_j)$ satisfy $a_{j+1} \le (1+2\delta_j)a_j + 2^{\,j+3-R_j}$ and are therefore bounded, say $\Phi(R_j) \le c_1 2^{-j}$.

Next, let $U_0 = R_0^2$ and $U_{j+1} = 2^{U_j}$. Then $R_j \le \sqrt{U_j}$ for all j: inductively, $R_{j+1} = 2^{256R_j} \le 2^{256\sqrt{U_j}} \le 2^{U_j/2}$ because $\sqrt{U_j} \ge 512$. Hence $\log^* R_j \le \log^* U_j = j + \log^* U_0$.

Finally let $r \ge R_0$ and let j be largest with $R_j \le r$. Then $\Phi(r) \le \Phi(R_j) \le c_1 2^{-j}$ and $\log^* r \le \log^* R_{j+1} \le j + 1 + \log^* U_0$. For $r < R_0$ use $\Phi \le 1$. □

Proof of the lower bound in Theorem 1.1. Let G be a pancyclic graph with n vertices and $n+h$ edges. It is connected, so its rank is $r = h+1$, and by Corollary 3.3

$$n - 2 \;\le\; \Phi(r)\, 2^r \;\le\; c\, 2^{\,r - \log^* r}.$$

The trivial bound $n - 2 \le 2^r - 1$ gives $n \le 2^{r+1} \le 2^{2^r}$, hence $\log^* n \le \log^* r + 2$. Taking logarithms, $h = r - 1 \ge \log(n-2) + \log^* r - \log c - 1 \ge \log n + \log^* n - \log c - 5$. □

4 The construction

Proposition 4.1. *For every $n \ge 3$ there is a pancyclic graph with n vertices and at most $n + \lfloor \log n \rfloor + \log^* n + 1$ edges.*

Proof. For $3 \le n \le 6$ take a cycle $u_0 u_1 \cdots u_{n-1}$, with the chord u_0u_2 if $n \ge 4$ and also the chord u_0u_3 if $n = 6$.

Let $n \ge 7$, and let $k \ge 2$ be the largest integer with $2^k + k + 1 \le n$, so that $n \le 2^{k+1} + k + 1$. Let H be a cycle of length n and let $Q \subseteq H$ be a path composed of consecutive segments $Q_0, \dots, Q_{k-1}$, where Q_i has $2^i + 1$ edges and joins v_i to v_{i+1}. Thus Q has $2^k + k - 1 \le n - 2$ edges. Add the chords $e_i = v_i v_{i+1}$ for $0 \le i < k$, and let N_j, the *necklace with j beads*, be the union of $Q_0, \dots, Q_{j-1}$ and $e_0, \dots, e_{j-1}$ (Figure 1).

A path from v_0 to v_j in N_j uses either Q_i or e_i for each $i < j$, and replacing Q_i by e_i shortens it by 2^i. So N_j has 2^j such paths, exactly one of each length from j to $2^j + j - 1$. Consequently a chord $f_j = v_0 v_j$ with $j \ge 2$ creates cycles of all lengths in $I_j = [\,j+1,\ 2^j + j\,]$, and the path $H - Q$ together with N_k contains cycles of all lengths in $[\,n - 2^k + 1,\ n\,]$.

Let $j_0 = k$ and $j_{i+1} = \lceil \log j_i \rceil$ as long as $j_i > 2$; this sequence decreases and ends with some $j_q = 2$. Add the chords $f_{j_0}, \dots, f_{j_q}$. Since $2^{j_{i+1}} \ge j_i$, the intervals $I_{j_{i+1}}$ and I_{j_i} overlap or are adjacent, and $I_2 = [3, 6]$; so we get all lengths in $[3,\ 2^k + k]$. Together with $[\,n - 2^k + 1,\ n\,]$ this is everything, except for the single length $2^k + k + 1$ in the case $n = 2^{k+1} + k + 1$, which is supplied by one more chord, joining v_k to the neighbour of v_0 on $H - Q$.

For an integer $j \ge 3$ we have $\log^* \lceil \log j \rceil = \log^* j - 1$, so $q = \log^* k - 1$. The number of chords is therefore at most $k + \log^* k + 1 \le \lfloor \log n \rfloor + \log^* n + 1$. □

5 The number of cycles

For the lower bound $\psi(r) > 2^{r-1}$, let $k = r - 1$ and replace every edge of a cycle of length k by a four-cycle in which the two ends of the edge are opposite vertices. The resulting graph has rank $k + 1 = r$, and its cycles are the k four-cycles and the 2^k cycles that use one half of each four-cycle; so $\psi(r) \ge 2^{r-1} + r - 1$. Horak and Kim [10] obtained $\psi(r) \ge 2^{r-1} + c\, r\, 2^{r/2}$.

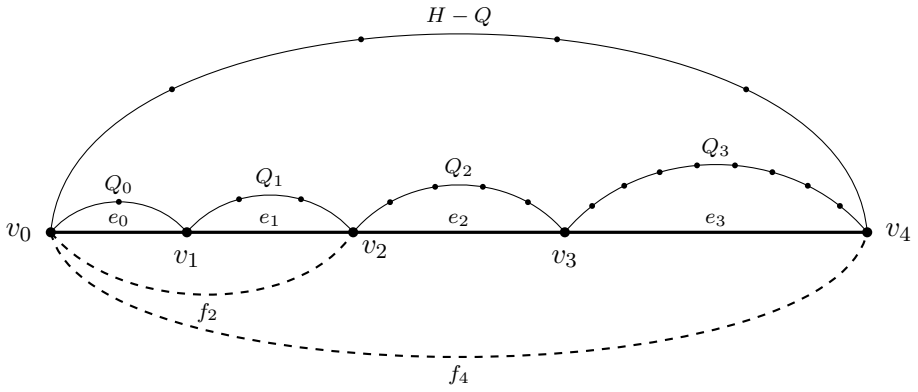


Figure 1: The construction for $n = 24$, where $k = 4$. The segments $Q_0, \dots, Q_3$ of the outer cycle H have $2, 3, 5, 9$ edges and are doubled by the chords $e_0, \dots, e_3$. The chord f_4 gives the cycle lengths $5, \dots, 20$, the rest $H - Q$ of the outer cycle gives $9, \dots, 24$, and f_2 gives $3, \dots, 6$.

Proof of Theorem 1.3. Let r be large and let G_0 be a graph of rank r with $\psi(r)$ cycles. The following operations change neither the rank nor the number of cycles: identifying a vertex of one component with a vertex of another; deleting a vertex of degree at most one; replacing a vertex of degree two and its two edges by a single edge. The last operation may create loops and parallel edges; a loop counts as a cycle of length one and two parallel edges form a cycle of length two. Applying the operations as long as possible, we obtain a connected multigraph K of rank r with $\psi(r)$ cycles and minimum degree at least 3, and hence with at most $2r - 2$ vertices.

Claim. If K' is obtained from K by deleting j edges and K' contains a cycle, then K' contains a cycle with at most $2\log r + 2j + 4$ edges.

Indeed, call $\sum_v \max(0, 3 - \deg v)$ the deficit of a multigraph; the deficit of K' is at most $2j$. Deleting a vertex of degree at most one does not increase the deficit, so after deleting such vertices as long as possible we are left with a multigraph K_1 of minimum degree at least two with at most $2j$ vertices of degree two. If a component of K_1 is a cycle, that cycle has at most $2j$ edges. Otherwise replace each vertex of degree two by an edge; the result K_2 has minimum degree at least 3 and at most $2r-2$ vertices. Let $d = \lceil \log(2r) \rceil$. If K_2 had no cycle with at most $2d$ edges, the vertices at distance at most d from a given vertex would be distinct, and there would be at least $3 \cdot 2^d - 2 > 2r$ of them. A cycle of K_2 with at most $2d$ edges comes from a cycle of K_1 with at most $2d + 2j \le 2\log r + 2j + 4$ edges. This proves the claim.

Now let $t = \lfloor (\log r)/100 \rfloor$. For $i = 1, \dots, t$ let C_i be a shortest cycle of $K - \{e_1, \dots, e_{i-1}\}$, which has positive rank, and let e_i be one of its edges. By the claim, $|C_i| \le 3\log r$. The cycles $C_1, \dots, C_t$ are linearly independent in $\mathcal{Z}(K)$, because $e_i \in C_i$ and $e_i \notin C_j$ for $j > i$.

Let G be the graph obtained from K by subdividing every edge twice. It is simple and connected, has rank r, and has $\psi(r)$ cycles. Let $W \subseteq G$ be the union of the cycles that correspond to $C_1, \dots, C_t$. Then $\beta(W) \ge t$, and W has at most $t \le (\log r)/100$ components and at most $9t \log r \le \sqrt{r}$ edges. By Lemma 2.1(a) and Theorem 1.2,

$$\psi(r) \;\le\; 2^r\big(p(G,W) + 2^{-t}\big) \;\le\; 2^r\Big(\frac{1}{2} + \frac{20}{\log^{(3)} r} + 2r^{-1/100}\Big) \;\le\; \Big(\frac{1}{2} + \frac{21}{\log^{(3)} r}\Big)2^r. \qquad \square$$

Remark 5.1. Theorem 1.3 is purely asymptotic. Its bound is below the trivial bound 2^r only

when $\log^{(3)} r > 42$, that is, for $r > 2^{2^{2^{42}}}$; it is below the bound $\frac{15}{16}2^r$ of [2] only when $\log^{(3)} r > 48$; and the threshold r_0 of Theorem 1.2 is not explicit. So for graphs of any moderate rank $\frac{15}{16}2^r$ remains the best bound known, and the theorem says nothing about the second-order term of $\psi(r)$ (see Section 8).

6 Proof of the forest estimate

In this section graphs may have loops and parallel edges. A loop adds two to the degree of its vertex and is a cycle of length one; two parallel edges form a cycle of length two. For a set A of vertices of G we write $G[A]$ for the induced subgraph, $G-A$ for $G[V(G)\setminus A]$, and δA for the set of edges with exactly one end in A. A *region* is a nonempty set A such that $G[A]$ is connected; it is *cyclic* if $G[A]$ contains a cycle. As before ξ is uniform on $\mathcal{Z}(G)$, and $\mathcal{O}_A$ denotes the event $\xi \cap \delta A = \emptyset$.

6.1 Two cuts share at most one bit

Lemma 6.1. *Let $Z \geq 0$ be a random variable with $\mu = \mathbb{E}Z > 0$ and $\mathbb{E}Z^2 \leq 2\mu^2 + K\mu$. Then $\mathbb{P}(Z = 0) \leq \frac{1}{2} + K/(4\mu)$.*

Proof. By the Cauchy–Schwarz inequality $\mu^2 \leq \mathbb{E}Z^2 \cdot \mathbb{P}(Z > 0)$, so $\mathbb{P}(Z = 0) \leq 1 - \mu/(2\mu + K) \leq \frac{1}{2} + K/(4\mu)$. □

For disjoint regions A, B of a connected graph G let $\ell(A, B)$ be the number of *links* between them: the edges with one end in A and the other in B, and the components of $G - (A \cup B)$ that are joined by an edge to A and by an edge to B.

Lemma 6.2 (Link formula)**.** *Let A and B be disjoint regions of a connected graph G. Then $\mathbb{P}(\mathcal{O}_A) = 2^{\mathrm{comp}(G-A)-|\delta A|} \geq 2^{-|\delta A|}$ and*

$$\mathbb{P}(\mathcal{O}_A \cap \mathcal{O}_B) = 2^{\ell(A,B)-1}\,\mathbb{P}(\mathcal{O}_A)\,\mathbb{P}(\mathcal{O}_B).$$

Conditionally on $\mathcal{O}_A$, the set $\xi \cap E(G[A])$ is uniform on $\mathcal{Z}(G[A])$; conditionally on $\mathcal{O}_A \cap \mathcal{O}_B$, the sets $\xi \cap E(G[A])$ and $\xi \cap E(G[B])$ are independent and uniform on $\mathcal{Z}(G[A])$ and $\mathcal{Z}(G[B])$.

Proof. The even subgraphs that avoid δA are the sums of an even subgraph of $G[A]$ and an even subgraph of $G - A$. So $\mathbb{P}(\mathcal{O}_A) = 2^{\beta(G[A])+\beta(G-A)-\beta(G)}$, and counting edges and vertices shows that the exponent is $\mathrm{comp}(G - A) - |\delta A|$. In the same way

$$\mathbb{P}(\mathcal{O}_A \cap \mathcal{O}_B) = 2^{\beta(G[A])+\beta(G[B])+\beta(G-A-B)-\beta(G)} = 2^{\mathrm{comp}(G-A-B)+1-|\delta A|-|\delta B|+e(A,B)},$$

where $e(A, B)$ is the number of edges between A and B. Let a, b and k be the numbers of components of $G - A - B$ that are joined to A only, to B only, and to both. Then $\mathrm{comp}(G - A - B) = a + b + k$, while $\mathrm{comp}(G - A) = a + 1$ and $\mathrm{comp}(G - B) = b + 1$, and the formula follows since $\ell(A, B) = k + e(A, B)$. The conditional laws are read off from the two direct sum decompositions. □

So two cut events are independent when there is one link, correlated by a factor of exactly two when there are two, and more strongly correlated only for pairs with at least three links. The next lemma says that such pairs are rare.

Lemma 6.3 (Exceptional partners)**.** *Let A be a region of a connected graph G, and let $B_1, \dots, B_k$ be pairwise disjoint regions, disjoint from A, with $\ell(A, B_i) \geq 3$ for all i. Then $k \leq |\delta A|$.*

Proof. Contract A to a vertex a and each B_i to a vertex b_i, and delete the loops that arise. This changes neither $d = |\delta A|$ nor the numbers $\ell(A, B_i)$, because the sets B_j with $j \neq i$ are connected and are contracted inside components of $G - (A \cup B_i)$. Let F be a spanning forest of $G - a$, and for every edge $e = au$ at a attach a new leaf τ_e to u. Call the resulting forest F^+ and its d new leaves *terminals*.

Fix i. Every link between a and b_i yields a path of F^+ from b_i to a terminal. For an edge $e = ab_i$ it is the edge $b_i\tau_e$. For a component D of $G - \{a, b_i\}$ joined to both, choose an edge $e = au$ with $u \in D$ and take the path of F^+ from b_i to τ_e; after its first vertex it stays in $D \cup \{\tau_e\}$. Paths that come from different links therefore start with different edges at b_i, and the union of any two of them is a path between two terminals. Let $F^\circ \subseteq F^+$ be the union of all paths between terminals. As $\ell(a, b_i) \geq 3$, the vertex b_i has degree at least 3 in F°. All leaves of F° are terminals, and a forest with at most d leaves has at most $d - 2$ vertices of degree at least 3. Hence $k \leq d - 2$. □

Lemma 6.4 (Many small cuts)**.** *Let $A_1, \dots, A_k$ be pairwise disjoint cyclic regions of a connected graph G with $|\delta A_i| \leq d$ for all i. Then*

$$\mathbb{P}(\xi \cap E(G[A_i]) \text{ is a forest for every } i) \;\leq\; \frac{1}{2} + \frac{(d+1)\,2^{d-1}}{k}.$$

Proof. Let E_i be the event that $\mathcal{O}_{A_i}$ holds and $\xi \cap E(G[A_i]) \neq \emptyset$. On E_i the set $\xi \cap E(G[A_i])$ is a nonempty even subgraph, hence not a forest, so it suffices to bound $\mathbb{P}(Z = 0)$ for $Z = \sum_i \mathbf{1}_{E_i}$. By Lemma 6.2, $p_i := \mathbb{P}(E_i) = \mathbb{P}(\mathcal{O}_{A_i})(1 - 2^{-\beta(G[A_i])}) \geq 2^{-d-1}$ and $\mathbb{P}(E_i \cap E_j) = 2^{\ell(A_i, A_j)-1} p_i p_j$ for $i \neq j$. If $\ell(A_i, A_j) \leq 2$ this is at most $2p_ip_j$. Otherwise we use $\mathbb{P}(E_i \cap E_j) \leq p_i$; by Lemma 6.3 this happens for at most d indices j, for each i. Hence $\mathbb{E}Z^2 \leq \mu + 2\mu^2 + d\mu$ with $\mu = \mathbb{E}Z \geq k\,2^{-d-1}$, and Lemma 6.1 gives the claim. □

6.2 Reduction to a subcubic graph

Lemma 6.5. *Let G and W be as in Theorem 1.2, with $m = |E(W)|$ and $c = \mathrm{comp}(W)$. There are a connected multigraph Γ with $\beta(\Gamma) = r$ and all degrees at most 3, and a set $P \subseteq V(\Gamma)$, such that*

(i) *every vertex outside P has degree 3, $|P| \leq 2m$, and $\Gamma[P]$ has at most c components;*

(ii) *$p(G, W) \leq \mathbb{P}(\xi_\Gamma \cap E(\Gamma - P)$ is a forest$)$, where ξ_Γ is uniform on $\mathcal{Z}(\Gamma)$.*

In particular $2r - 2 \leq |V(\Gamma)| \leq 2r + 2m$.

Proof. Splitting. Replace every vertex v of degree $d \geq 4$ by a path $v^1 v^2 \cdots v^{d-2}$ and distribute the d edges at v along it in some order: two to v^1, one to each of $v^2, \dots, v^{d-3}$, and two to v^{d-2}. If $v \in V(W)$, let the edges of W at v come first in this order, so that they end in an initial segment $v^1, \dots, v^i$ of the path with $i < \deg_W(v)$. Let G' be the new graph; its degrees are at most 3. Let P consist of these initial segments and of the vertices of W that were not split. Then $|P| \leq \sum_v \deg_W(v) = 2m$, every edge of W joins two vertices of P, and $G'[P]$ has at most c components.

Contracting the new path edges maps $\mathcal{Z}(G')$ bijectively onto $\mathcal{Z}(G)$, so we may take $\xi_G = \xi_{G'} \cap E(G)$. Every edge of $G' - P$ is a new path edge or an edge of $E(G) \setminus E(W)$. If $\xi_G \setminus E(W)$ is a forest, then so is its union with the set of all new path edges, because contracting these edges, which form a forest, gives back $\xi_G \setminus E(W)$. Hence $p(G,W) \le \mathbb{P}(\xi_{G'} \cap E(G' - P)$ is a forest$)$.

Cleaning. As long as possible, delete a vertex of degree one outside P, or replace a vertex of degree two outside P and its two edges by a single edge. An even subgraph avoids a pendant edge and contains both or none of the two edges at a vertex of degree two, so these operations induce bijections of the cycle spaces. Let Γ be the final multigraph. Each edge of $\Gamma - P$ is a path of G' all of whose vertices lie outside P; so if ξ_Γ contains a cycle of $\Gamma - P$, then the corresponding $\xi_{G'}$ contains a cycle of $G' - P$. This proves (ii).

Edges with both ends in P are never removed, so the vertices of P keep degree at least two, and the number of components of the graph induced on P does not increase. The vertices outside P now have degree exactly 3. Finally $3|V(\Gamma)| - |P| \le 2|E(\Gamma)| \le 3|V(\Gamma)|$ and $|E(\Gamma)| = |V(\Gamma)| + r - 1$ give the bounds on $|V(\Gamma)|$. □

6.3 Parameters, and removing the short cycles

From now on Γ and P are as in Lemma 6.5, $\xi = \xi_\Gamma$, and r is large. All $o(\cdot)$ and $O(\cdot)$ refer to $r \to \infty$ and are uniform in G and W. Put

$$x = \log r, \qquad y = \log x, \qquad z = \log y,$$
$$D = \lfloor x/10 \rfloor, \qquad L = \lfloor x\sqrt{y} \rfloor, \qquad s = \lfloor (D-1)/2 \rfloor, \qquad \varepsilon = 20/z.$$

Recall that $|P| \le 2\sqrt{r}$ and $c \le x/100$. Let $\mathcal{F}$ be the event that $\xi \cap E(\Gamma - P)$ is a forest. We assume that

$$\mathbb{P}(\mathcal{F}) > \tfrac{1}{2} + \varepsilon \tag{2}$$

and derive a contradiction; by Lemma 6.5(ii) this proves Theorem 1.2.

Let $\mathcal{Q}$ be a maximal family of pairwise disjoint cycles of $\Gamma - P$ of length at most D, loops and pairs of parallel edges included, and let S be the union of their vertex sets. For $Q \in \mathcal{Q}$ the set $V(Q)$ is a cyclic region of Γ with $|\delta V(Q)| \le |Q| \le D$, because its vertices have degree 3. On $\mathcal{F}$ the set $\xi \cap E(\Gamma[V(Q)])$ is a forest for every Q, so Lemma 6.4 and (2) give $|\mathcal{Q}| < (D+1)2^{D-1}/\varepsilon \le \sqrt{r}$. Hence $|S| \le D\sqrt{r}$.

Let $J = \Gamma - (P \cup S)$. By the maximality of $\mathcal{Q}$, J is a simple graph of girth greater than D. It has $v = 2r + O(\sqrt{r}\log r)$ vertices, each of degree 3 in Γ, so its *deficit* satisfies

$$d_J := \sum_{u \in V(J)} (3 - \deg_J u) \;\le\; 3(|P| + |S|) = O(\sqrt{r}\log r). \tag{3}$$

6.4 Cycles of moderate length

A walk $u_0 u_1 \cdots u_\ell$ is *non-backtracking* if $u_{i+1} \ne u_{i-1}$ for $0 < i < \ell$. Let $N_\ell(u,w)$ be the number of non-backtracking walks of length ℓ from u to w in J.

Lemma 6.6. *If $\ell > s$, then $N_\ell(u,w) \le 3 \cdot 2^{\ell-s-1}$ for all u, w. Consequently, for every vertex u of J, the cycles C of J through u with $|C| \le L$ satisfy $\sum_C 2^{-|C|} \le L\, 2^{-s}$.*

Proof. There are at most $3 \cdot 2^{\ell-s-1}$ ways to choose the first $\ell - s$ steps. The last s steps form a non-backtracking walk of length s between two given vertices. Such a walk is a path, because

s is less than the girth, and there is at most one such path, because two of them would give a cycle of length at most $2s < D$. For the second statement, a cycle of length ℓ through u can be traversed from u in two directions, so there are at most $\frac{1}{2}N_\ell(u,u)$ of them, and $\ell > D > s$. $\square$

Let B be the non-backtracking matrix of J. Its rows and columns are indexed by the directed edges $(i \to j)$ of J, two for each edge, and the entry in row $(i \to j)$ and column $(j' \to k)$ is 1 if $j' = j$ and $k \neq i$, and 0 otherwise.

Lemma 6.7. *For every even $\ell \geq 2$,*

$$\operatorname{tr} B^\ell \;\geq\; \bigl(2 - d_J/v\bigr)^\ell - 3v\,2^{\ell/2}.$$

Proof. Let A be the adjacency matrix of J and Δ the diagonal matrix of its degrees, and put $M(\lambda) = \lambda^2 I - \lambda A + \Delta - I$.

(1) *If $M(\lambda)g = 0$ for a real $\lambda > 1$ and a vector $g \neq 0$, then λ is an eigenvalue of B.* A direct computation shows that $f(i \to j) = \lambda g(j) - g(i)$ satisfies $Bf = \lambda f$; and $f = 0$ would give $g(i) = \lambda g(j) = \lambda^2 g(i)$ along every edge, hence $g = 0$ (at an isolated vertex, $M(\lambda)g = 0$ already says $(\lambda^2 - 1)g = 0$).

(2) *If $Bf = \lambda f$ with $f \neq 0$ and $\lambda^2 \neq 1$, then $M(\lambda)g = 0$ for $g(j) = \sum_{k \sim j} f(j \to k)$, and $g \neq 0$.* The eigenvalue equation at $(i \to j)$ and at $(j \to i)$ reads $\lambda f(i \to j) = g(j) - f(j \to i)$ and $\lambda f(j \to i) = g(i) - f(i \to j)$. Eliminating $f(j \to i)$ gives $(\lambda^2 - 1)f(i \to j) = \lambda g(j) - g(i)$, and summing this over the neighbours j of i gives $M(\lambda)g = 0$.

(3) *B has a real eigenvalue $\lambda_1 \geq 2 - d_J/v$.* The matrix $M(2) = 2(\Delta - A) + (3I - \Delta)$ is positive semidefinite, while $\mathbf{1}^\top M(\lambda)\mathbf{1} = (\lambda - 1)(v\lambda - 2v + d_J)$ vanishes at $\lambda^* = 2 - d_J/v > 1$. The smallest eigenvalue of the symmetric matrix $M(\lambda)$ is continuous in λ, at most 0 at λ^* and at least 0 at 2. So $M(\lambda_1)$ is singular for some $\lambda_1 \in [\lambda^*, 2]$, and (1) applies.

(4) *Every non-real eigenvalue λ of B satisfies $|\lambda| \leq \sqrt{2}$.* With g as in (2), $g^* M(\lambda) g = 0$ is a quadratic equation $a\lambda^2 - b\lambda + c' = 0$ with real coefficients $a = g^*g > 0$, $b = g^*Ag$ and $c' = g^*(\Delta - I)g \leq 2a$. Its roots are λ and $\bar{\lambda}$, so $|\lambda|^2 = c'/a \leq 2$.

Now $\operatorname{tr} B^\ell$ is the sum of the ℓth powers of the $2|E(J)| \leq 3v$ eigenvalues of B. For even ℓ the real eigenvalues contribute nonnegative terms, one of which is at least $(2 - d_J/v)^\ell$, and the non-real ones contribute at least $-3v\,2^{\ell/2}$ in total. $\square$

Facts (2) and (4) are part of the Ihara–Bass theory [3, 12]; the use of the test vector $\mathbf{1}$ in (3), which makes the argument work without any assumption of regularity or expansion, is from [11].

Lemma 6.8 (Cycle weight)**.** *Let $\mathcal{C}$ be the set of cycles of J of length at most L. Then*

$$V := \sum_{C \in \mathcal{C}} 2^{-|C|} \;\geq\; \bigl(\tfrac{1}{8} - o(1)\bigr) \ln y.$$

Proof. $\operatorname{tr} B^\ell$ is the number of closed walks $u_0 u_1 \cdots u_\ell = u_0$ in J with $u_{i+1} \neq u_{i-1}$ for all i modulo ℓ. A cycle of length ℓ gives 2ℓ of them; let n_ℓ be the number of the others. Then

$$T := \sum_{\ell \leq L} \frac{\operatorname{tr} B^\ell}{2\ell\, 2^\ell} = V + \sum_{\ell \leq L} \frac{n_\ell}{2\ell\, 2^\ell}.$$

Consider a walk counted by n_ℓ, and let j be the least index with $u_j \in \{u_0, \ldots, u_{j-1}\}$, say $u_j = u_i$. Then $j < \ell$, the walk $Q = u_i u_{i+1} \cdots u_j$ is a cycle, of length q say, and $u_j u_{j+1} \cdots u_\ell u_1 \cdots u_i$ is

a non-backtracking closed walk of length $\ell - q > 0$ from u_i to itself. It contains a cycle, so $\ell - q > D > s$. The original walk is determined by Q, by the vertex u_i and a direction on Q ($2q$ choices), by the complementary walk (at most $3 \cdot 2^{\ell-q-s-1}$ choices, by Lemma 6.6), and by the position of u_0 on the latter (at most ℓ choices). Hence

$$\sum_{\ell \le L} \frac{n_\ell}{2\ell\, 2^\ell} \;\le\; \sum_{\ell \le L} \sum_{Q \in \mathcal{C}} \tfrac{3}{2}\, |Q|\, 2^{-|Q|-s} \;\le\; \tfrac{3}{2}\, L^2\, 2^{-s}\, V = o(V),$$

because $2^{-s} \le 8\, r^{-1/20}$ and $L^2 \le x^2 y$.

On the other hand, for even ℓ with $6x \le \ell \le L$, Lemma 6.7 and (3) give

$$2^{-\ell} \operatorname{tr} B^\ell \;\ge\; \Big(1 - \frac{d_J}{2v}\Big)^\ell - 3v\, 2^{-\ell/2} \;\ge\; 1 - \frac{L\, d_J}{2v} - 3v\, r^{-3} = 1 - o(1).$$

All terms of T are nonnegative, so $T \ge (1 - o(1)) \sum 1/(2\ell)$ over the even ℓ in $[6x, L]$, and this sum is $(\frac{1}{4} - o(1)) \ln(L/6x) = (\frac{1}{8} - o(1)) \ln y$. Therefore $V = (1 - o(1))T \ge (\frac{1}{8} - o(1)) \ln y$. □

6.5 The second moment

For $C \in \mathcal{C}$ let I_C be the event that C is a component of ξ. As the vertices of C have degree 3 in Γ, this is the event $E(C) \subseteq \xi$; it is also the event that $\mathcal{O}_{V(C)}$ holds and $\xi \cap E(\Gamma[V(C)]) = E(C)$. Put

$$a_C = 2^{-|C|}, \qquad p_C = \mathbb{P}(I_C), \qquad X_C = \frac{a_C}{p_C}\, \mathbf{1}_{I_C}, \qquad Z = \sum_{C \in \mathcal{C}} X_C.$$

The event I_C is given by $|C|$ linear conditions on $\xi \in \mathcal{Z}(\Gamma)$, which are satisfied by $\xi = C$; so $p_C \ge a_C$. Hence $0 \le X_C \le 1$, $\mathbb{E}X_C = a_C$ and $\mu := \mathbb{E}Z = V$. On $\mathcal{F}$ no cycle of $J \subseteq \Gamma - P$ is contained in ξ, so $Z = 0$, and by (2)

$$\mathbb{P}(Z = 0) > \tfrac{1}{2} + \varepsilon. \tag{4}$$

Let $C \ne C'$ be in $\mathcal{C}$. If they share a vertex, then $I_C \cap I_{C'} = \emptyset$. If they are disjoint, Lemma 6.2, applied to the regions $V(C)$ and $V(C')$ of Γ, gives $\mathbb{P}(I_C \cap I_{C'}) = 2^{\ell(C,C')-1} p_C p_{C'}$ and therefore

$$\mathbb{E}(X_C X_{C'}) = a_C a_{C'}\, 2^{\ell(C,C')-1}, \tag{5}$$

where $\ell(C, C')$ is the number of links between $V(C)$ and $V(C')$ in Γ.

Lemma 6.9 (Few links)**.** *$\ell(C, C') \le c + o(x)$ for all disjoint $C, C' \in \mathcal{C}$.*

Proof. Let $U = V(C) \cup V(C')$. Every vertex of U lies on exactly one edge outside its own cycle, so at most $2L$ edges leave U. At most c components of $\Gamma - U$ meet P, because $P \cap U = \emptyset$ and so each component of $\Gamma[P]$ lies in one of them. Let Y run over the other components and let d_Y be the number of edges between Y and U; then $\sum_Y d_Y \le 2L$. Put $t_0 = \lfloor y/2 \rfloor$.

Large cuts. At most $2L/t_0$ components have $d_Y > t_0$.

Small cuts, cyclic. The components Y with $d_Y \le t_0$ that are cyclic are pairwise disjoint cyclic regions inside $\Gamma - P$ with $|\delta Y| \le t_0$. On $\mathcal{F}$ each $\xi \cap E(\Gamma[Y])$ is a forest, so by Lemma 6.4 and (2) there are fewer than $(t_0 + 1)2^{t_0-1}/\varepsilon$ of them.

Small cuts, trees. Let Y induce a tree, with $d_Y \le t_0$. No $Q \in \mathcal{Q}$ meets Y, since Q is a connected subgraph of $\Gamma - U$ and would lie in Y. Hence $Y \subseteq V(J)$. As all degrees are 3, we have $3|Y| = 2(|Y| - 1) + d_Y$, that is, $|Y| = d_Y - 2$. So if Y is joined to both cycles, there is a

path R from C to C' through Y with at most t_0 edges. For a link that is an edge, let R be that edge. The paths R obtained in this way lie in J and are pairwise disjoint. Divide each of C and C' into at most $8L/D$ blocks of at most $D/4$ consecutive vertices. If two of the paths had their ends in the same block of C and in the same block of C', they would form, together with two arcs inside these blocks, a cycle of J of length less than $2t_0 + D/2 \le D$. So there are at most $(8L/D)^2$ such paths.

Altogether

$$\ell(C,C') \;\le\; c + \frac{2L}{t_0} + \frac{(t_0+1)2^{t_0-1}}{\varepsilon} + \Big(\frac{8L}{D}\Big)^2 = c + O\Big(\frac{x}{\sqrt{y}}\Big) + O(yz\sqrt{x}) + O(y) = c + o(x). \;\square$$

Proof of Theorem 1.2. Call a disjoint pair $C, C' \in \mathcal{C}$ *exceptional* if $\ell(C,C') \ge 3$. Fix C, and let $\mathcal{E}$ be a maximal family of pairwise disjoint exceptional partners of C. By Lemma 6.3, $|\mathcal{E}| \le |\delta V(C)| \le L$, so the union U_C of their vertex sets has at most L^2 vertices, and every exceptional partner of C meets U_C. By Lemma 6.6, the sum of $a_{C'}$ over all $C' \in \mathcal{C}$ that meet U_C is at most $L^3 2^{-s}$. With (5) and Lemma 6.9,

$$\sum_{C' \text{ exceptional}} \mathbb{E}(X_C X_{C'}) \;\le\; a_C\, 2^{c+o(x)} L^3\, 2^{-s} = a_C\, 2^{\,c-x/20+o(x)} = o(a_C),$$

because $c \le x/100$. For the other disjoint pairs $\mathbb{E}(X_C X_{C'}) \le 2a_C a_{C'}$, intersecting pairs contribute nothing, and $\mathbb{E}X_C^2 \le \mathbb{E}X_C = a_C$. Summing over C we get $\mathbb{E}Z^2 \le 2\mu^2 + (1+o(1))\mu$. By Lemmas 6.1 and 6.8,

$$\mathbb{P}(Z=0) \;\le\; \frac{1}{2} + \frac{1+o(1)}{4\mu} \;\le\; \frac{1}{2} + \frac{2+o(1)}{\ln y} \;<\; \frac{1}{2} + \frac{20}{z}$$

for large r, since $\ln y = z \ln 2$. This contradicts (4). $\square$

Remark 6.10. The assumption $|E(W)| \le \sqrt{r}$ entered only through $|P| \le 2\sqrt{r}$ in (3), where all that is needed is $L\, d_J = o(v)$; the proof therefore works unchanged for $|E(W)| \le r/\log^2 r$. Similarly the constant 20 can be replaced by any constant larger than $2/\ln 2$. We have kept the statement of [11], which is the one that has been formalised.

7 Exact and nearly exact counting

By Lemma 2.1 the cycles of G have at most $2^{r-t}(pm + 1 - p)$ lengths, and Theorem 1.2 says that $p \le \frac{1}{2} + o(1)$ when r is large. Here we look at the other end of the scale. When is nothing lost, and what do graphs look like that lose little?

7.1 Two terminals

Proposition 7.1. *Let N be a connected graph and let $s \ne t$ be two of its vertices. Then N has at most $2^{\beta(N)}$ paths from s to t. If it has exactly $2^{\beta(N)}$, then $N - E(A)$ is a forest for every path A from s to t, and every such path has at least $\beta(N)$ edges.*

Proof. A path from s to t is an $\{s,t\}$-join, and the joins form a coset of $\mathcal{Z}(N)$, so there are $2^{\beta(N)}$ of them. Suppose that all joins are paths, and let A be one of them. If $N - E(A)$ contained a cycle C, then $A \cup C = A + C$ would be a join that is not a path. So $N - E(A)$ is a forest, and since deleting an edge lowers the cycle rank by at most one, $\beta(N) \le |A|$. $\square$

Equivalently, an edge f of a graph G lies on at most $2^{\beta(G)-1}$ cycles, and if it lies on that many, then each of them has at least $\beta(G)$ edges. Equality in Proposition 7.1 holds for the necklace N_j of Section 4 between v_0 and v_j, for K_4 minus an edge between its two vertices of degree two, and for $K_{3,3}$ minus an edge between the ends of the deleted edge.

This is the reason for the shape of the construction. The necklace N_k turns k units of rank into 2^k path lengths, the most that k units can give. But by Proposition 7.1 no gadget that does this has a path with fewer than k edges between its terminals. So the cycles through N_k have length greater than k, and the lengths $3, \dots, k$ have to be produced elsewhere: by a necklace with about $\log k$ beads, whose own short lengths require a third one, and so on. Relative to each necklace N_j that is closed by a chord f_j, taken together with f_j and the chords of the smaller necklaces, the rest of the graph is attached at the two vertices v_0 and v_j only, so that half of the words cannot be used (Remark 2.2). These $\log^* k$ halvings are the term $\log^* n$ of Theorem 1.1.

7.2 No waste

Call a word *wasted* if it is not the outside part of any cycle. In Lemma 2.1 nothing is lost only if no word is wasted, and this cannot go on for long.

Proposition 7.2. *Let G be a connected graph and let $W \subseteq G$ be a subgraph that contains a cycle. Suppose that every coset of $\mathcal{Z}(W)$ in $\mathcal{Z}(G)$ contains a cycle. Then every nonempty even subgraph of $G - E(W)$ is a cycle, and*

$$\beta(G) - \beta(W) \;\le\; |V(W)| + 3.$$

Proof. Let z be a nonempty even subgraph of $G - E(W)$. All members of the coset $z + \mathcal{Z}(W)$ contain z, one of them is a cycle, and a cycle that contains a nonempty even subgraph is equal to it. So z is a cycle. In particular the graph $G' = G - E(W)$, on the vertex set $V(G)$, has no two edge-disjoint cycles, and by Lemma 7.3 below $|E(G')| \le |V(G)| + 3$. Since $\beta(G) = |E(G)| - |V(G)| + 1$ and $\beta(W) \ge |E(W)| - |V(W)| + 1$, we get $\beta(G) - \beta(W) \le |E(G')| - |V(G)| + |V(W)| \le |V(W)| + 3$. □

Lemma 7.3 (Erdős and Pósa [7])**.** *A multigraph with n vertices and at least $n+4$ edges contains two edge-disjoint cycles.*

Proof. We include the short proof. Let H be a counterexample with $|V(H)| + |E(H)|$ minimal, and let n be its number of vertices. Then H has exactly $n + 4$ edges and no vertex of degree at most one. It has no vertex of degree two either: such a vertex could be replaced, together with its two edges, by a single edge, or it carries a loop, and the rest of H contains a second cycle. So $2(n + 4) \ge 3n$, that is, $n \le 8$. Let C be a shortest cycle of H. Since $H - E(C)$ is a forest, $n + 4 - |C| \le n - 1$, so $|C| \ge 5$. Thus H is a simple graph of minimum degree at least 3 and girth at least 5, and such a graph has at least $1 + 3 + 6 = 10$ vertices, a contradiction. □

The first conclusion of Proposition 7.2 is the graphic case of the fact that a binary code in which every nonzero word is minimal is intersecting [1, §2]. The second is sharp up to the additive constant. If W is a cycle $w_1 \cdots w_m$ and G consists of W and a path of length two from w_i to w_{i+1} for every i modulo m, then no word is wasted and $\beta(G) - \beta(W) = |V(W)|$. If G is K_5 minus an edge and W is the triangle on its three vertices of degree four, then no word is wasted and $\beta(G) - \beta(W) = |V(W)| + 1$. An exhaustive search found nothing better for a triangle W in graphs with at most eight vertices.

So the trivial bound of $m\,2^{r-t}$ cycle lengths can be exact for at most $m+3$ ranks beyond W, while by Theorem 1.2 it is off by a factor of almost two as soon as $r \geq m^2$, for large m and W as in that theorem.

7.3 A defect budget

The last statement of this section is a stability version of the same picture: a graph that comes close to the trivial bound must be built, at almost every step, the way the necklace is built.

Let G contain a cycle of each length $3, \dots, L$, and let $\Theta = (L-2)\,2^{-\beta(G)}$. A *greedy sequence* is a sequence of subgraphs $W_1 \subset W_2 \subset \cdots \subset W_Q$ of G in which W_1 is a triangle, $W_{q+1} = W_q \cup C_q$ for a cycle C_q of G whose length ℓ_q is the least integer ≥ 3 that is not the length of a cycle of W_q, and W_Q is the first term that contains cycles of all lengths $3, \dots, L$. Write $m_q = |E(W_q)|$ and $t_q = \beta(W_q)$, and let

$$\rho_q = \frac{m_q+2}{2^{t_q}}, \qquad \kappa_q = 1 - \frac{m_{q+1}-m_q}{m_q+2}, \qquad j_q = t_{q+1} - t_q - 1.$$

Proposition 7.4 (Defect budget)**.** *For every greedy sequence:*

(i) $0 < \kappa_q < 1$, $j_q \geq 0$ *and* $\rho_{q+1} = (1 - \frac{1}{2}\kappa_q)\,2^{-j_q}\rho_q$ *for* $1 \leq q < Q$;

(ii) $\Theta < \rho_q$ *for all* q, *and therefore* $\sum_{q=a}^{Q-1}(\frac{1}{2}\kappa_q + j_q \ln 2) < \ln(\rho_a/\Theta)$ *for every* a;

(iii) *if* $j_q = 0$ *and* $\kappa_q \leq \eta$, *then* C_q *has at least* $(1-\eta)(m_q+2)$ *edges outside* W_q *and fewer than* $\eta(m_q+2)$ *edges in* W_q, *and* W_q *has cycles of at least* $(1-\eta)(m_q+2)-3$ *different lengths.*

Proof. (i) Let $P_q = E(C_q) \setminus E(W_q)$, so that $|P_q| = m_{q+1} - m_q \geq 1$. The graph W_q has cycles of all lengths $3, \dots, \ell_q - 1$ and no cycle longer than m_q. Hence $|P_q| \leq \ell_q \leq m_q + 1$ and $0 < \kappa_q < 1$. Also $j_q \geq 0$, because $C_q \in \mathcal{Z}(W_{q+1}) \setminus \mathcal{Z}(W_q)$, and the identity restates $m_{q+1} + 2 = (2-\kappa_q)(m_q+2)$.

(ii) By Lemma 2.1(b) with $W = W_q$ and $p \leq 1$, the cycles of G have at most $m_q 2^{\beta(G)-t_q}$ lengths, so $\Theta \leq m_q 2^{-t_q} < \rho_q$. By (i), $\rho_Q \leq \rho_a \exp\bigl(-\sum_{q=a}^{Q-1}(\frac{1}{2}\kappa_q + j_q \ln 2)\bigr)$.

(iii) Here $|P_q| = (1-\kappa_q)(m_q+2)$, the number of edges of C_q in W_q is $\ell_q - |P_q| \leq m_q + 1 - |P_q|$, and W_q has cycles of at least $\ell_q - 3 \geq |P_q| - 3$ lengths. □

Call a step *perfect* if $j_q = 0$ and $\kappa_q \leq \eta$. By (ii), if G comes within a factor K of the trivial bound $\Theta < \rho_a$ relative to W_a, then all but $(2/\eta)\ln K$ of the steps after W_a are perfect. By (iii), a perfect step roughly doubles the graph by a new cycle that closes through a part of W_q of relative size less than η, at a moment when almost every integer up to m_q is the length of a cycle of W_q; and if W_q is connected, $j_q = 0$ means that the new edges form a single path or cycle that meets W_q in at most two vertices. These are the features of a bead of the necklace. In this coarse sense the shape of the construction is forced between any two halvings, and Theorem 1.2 forces the halvings.

8 Problems

Error terms. By Remark 2.2 the left side of (1) can be as large as $\frac{1}{2} + 2^{1-r}$, and Horak and Kim [10] proved that $\psi(r) \geq 2^{r-1} + c\,r\,2^{r/2}$ for some $c > 0$. The upper bounds of Theorems 1.2 and 1.3 are very far from this. In the proof, the gain μ comes from cycles with between $6\log r$ and $\log r\sqrt{\log\log r}$ edges. Each length ℓ contributes about $1/(2\ell)$, and the upper end of the range is dictated by Lemma 6.9; so μ is only of order $\log^{(3)} r$.

Problem 8.1. Is the error term in (1) at most r^{-c} for some $c > 0$? Is $\psi(r) \le 2^{r-1} + 2^{(1-c)r}$?

Between exactness and the forest estimate. Proposition 7.2 and Theorem 1.2 are an exact and an asymptotic statement about the same thing: no word is wasted for at most $|V(W)| + 3$ ranks beyond W, and about half of the words fail to be forests once $r \ge |E(W)|^2$, or already once $r \ge |E(W)| \log^2 r$ by Remark 6.10. We do not know a proof of the factor $\frac{1}{2}$ that avoids the second moment computation of Section 6, except in special situations such as Remark 2.2. A robust form of Proposition 7.1 would be a step in that direction.

Problem 8.2. Let $\varepsilon > 0$. If a connected graph N has at least $(\frac{1}{2} + \varepsilon)2^{\beta(N)}$ paths between two vertices s and t, must each of these paths have at least $c(\varepsilon)\beta(N)$ edges?

Problem 8.3. Can the constant 3 in Proposition 7.2 be replaced by 1?

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